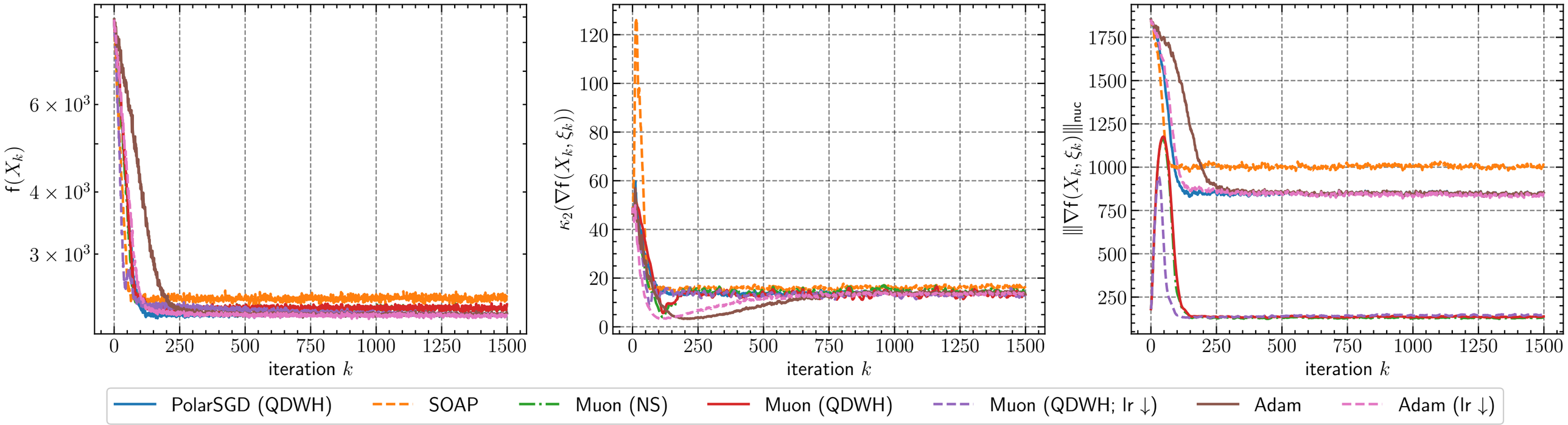
$f(X_k)$
6×10^3
4×10^3
3×10^3
iteration k
$\kappa_2(\nabla f(X_k, \xi_k))$
iteration k
$\|\nabla f(X_k, \xi_k)\|_{\text{nuc}}$
iteration k
PolarSGD (QDWH)
SOAP
Muon (NS)
Muon (QDWH)
Muon (QDWH; lr ↓)
Adam
Adam (lr ↓)